

Roll No.

TBT-702(a)

**B. TECH. (BIOTECHNOLOGY)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
MICROBIAL TECHNOLOGY**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Illustrate the production of recombinant insulin. (CO1)

OR

(b) Define microbial polysaccharides. Explain the production of xanthan gum. (CO1)

2. (a) Implement the knowledge of recombinant DNA technology in production of hGH. (CO3, CO1)

OR

(b) State the role of microorganisms in agriculture and pharmaceutical industries. (CO3, CO1)

3. (a) List the applications of dextran and xanthan gum. (CO1, CO2)

OR

(b) What are inclusion bodies ? Discuss the formation and prevention of inclusion bodies. (CO1, CO2)

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(2)

4. (a) Discuss the use of yeast vector in vaccine production. _____ (CO2)

OR

- (b) Explain the biosynthesis of Polyhydroxyalkanoate (PHA). (CO2)

5. (a) Write names of *five* Gram positive and *five* Gram negative bacteria.

(CO1)

OR

- (b) What do you understand by the term gene expression ? Define the process of post-transcriptional modification. (CO1)